

6.4 Prototype

پروٹوٹائپ (Prototype) کسی پراڈکٹ، سروس، ویب سائٹ یا ایپ کا ابتدائی نمونہ ہوتا ہے جو دکھاتا ہے کہ خیال کیسا نظر...

CS XII

Chapter 6: Entrepreneurship in the Digital Age

Lecture 05 of 16

★ Golden

6.4 Prototype ★ Golden

LEARN IT

A **prototype** is an **early model** of a product, service, website, or app. It shows **how the idea may look or work**. It is **NOT the final product**. It is made to **test the idea and collect feedback**.

Think of the clay model an architect makes before building a house, or the sample *kameez* a tailor stitches before making 50 for a shop. The model lets people say "make this bigger" or "move that here" before real money is spent.

Why is a prototype useful?

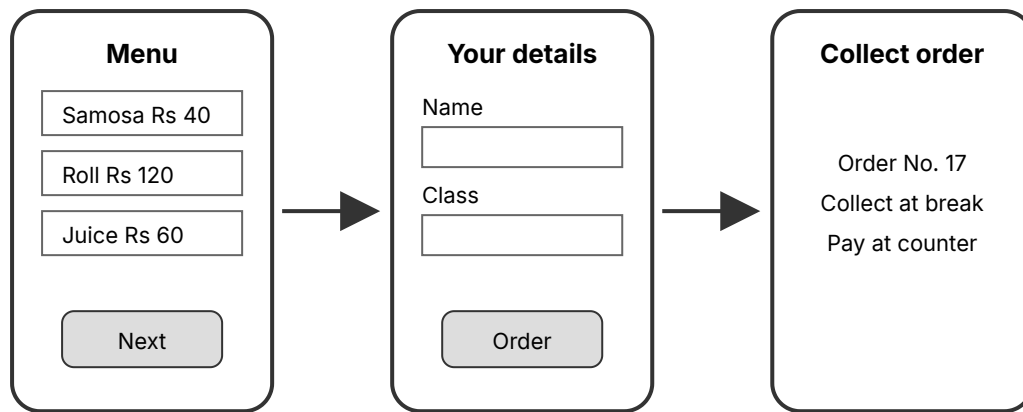
A prototype helps students and entrepreneurs **explain their idea clearly**, because users can **see something** instead of only listening to words.

- **It makes the idea visible.**
- **It helps the team find mistakes early.** Users can point out if a button is hard to find, if a screen has too much text, or if a feature is missing.
- **It saves time and money** before final coding begins. Fixing a pencil sketch takes a minute; fixing finished code can take days.

What form can a prototype take?

A prototype may be a **paper sketch**, a **cardboard model**, a **clickable PowerPoint**, a **Canva design**, a **Figma wireframe**, a **simple website layout**, or a **sample service process chart**. A *wireframe* is a simple outline drawing of a screen that shows where things go (boxes, buttons, text) without colours or pictures.

For example, for a **canteen pre-order system**, students can just **draw the screens on paper**.



A paper prototype of a canteen pre-order system: three hand-drawn screens joined by arrows (select item → enter name → collect order).

WORKED EXAMPLE

Worked example: how a paper prototype finds a mistake early. Bilal's team in a Nawabshah college draws the three screens above on A4 paper.

1. They hand screen 1 to a classmate and say, "Order a samosa."
2. The classmate looks for a quantity box and cannot find one. She asks, "What if I want two?"
3. The team notes: **feature missing: quantity**.
4. They add a small "Qty: []" box with a pencil in 30 seconds.

If they had already coded an app, adding quantity would have meant changing the database, the screen and the order list. The prototype **saved time and money**.

REMEMBER IT

Prototype = "practice model". It is like a rough draft of an essay: you show it, get corrections, and rewrite it before the final copy.

COMMON MISTAKE

Mistake: Thinking a prototype must work or must look beautiful.

Fix: A prototype only has to make the idea **visible** so users can react. A pencil sketch is a valid prototype.

پروٹوٹائپ (Prototype) کسی پراڈکٹ، سروس، ویب سائٹ یا ایپ کا ابتدائی نمونہ ہوتا ہے جو دکھاتا ہے کہ خیال کیسا نظر آئے گا یا کیسے کام کرے گا۔ یہ حتمی پراڈکٹ نہیں ہوتا بلکہ خیال کو آزمانے اور رائے (Feedback) لینے کے لیے بنایا جاتا ہے۔ یہ کاغذ کا خاکہ، گتے کا ماڈل، پاورپوائنٹ یا فگما (Figma) ڈیزائن بھی ہو سکتا ہے، اور اس سے غلطیاں جلد پکڑی جاتی ہیں اور وقت اور پیسہ بچتا ہے۔

EXAM TIP

Expect "Define prototype and write its benefits" (3–5 marks). Write: (1) definition with the words *early model, not the final product, test the idea and collect feedback*; (2) three benefits: makes the idea visible, finds mistakes early, saves time and money; (3) forms it can take (paper sketch, cardboard model, clickable PowerPoint, Canva, Figma wireframe); (4) the canteen paper-screens example.

CHECK YOURSELF

1. Is a prototype the final product?
2. Name four forms a prototype can take.
3. Give two kinds of problems users can point out by looking at a prototype.

Answers: 1. No. It is an early model made to test the idea and collect feedback. 2. Any four: paper sketch, cardboard model, clickable PowerPoint, Canva design, Figma wireframe, simple website layout, sample service process chart. 3. Any two: a button is hard to find, a screen has too much text, a feature is missing.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A prototype is:

- a. An early model of how it may look/work — not the final product
- b. The finished mass-produced item
- c. A database backup
- d. A pie of revenue

MCQ 2. You build it to:

- a. Learn, not to impress investors first
- b. Win a logo contest only
- c. Replace users
- d. Skip tests

MCQ 3. This is ★: define, purpose, what it is NOT.

- a. The 5-mark
- b. Only draw Figma pretty
- c. Only Python
- d. Only OSI

MCQ 4. A paper phone screen is still a:

- a. Prototype (low fidelity)
- b. MVP already charged
- c. Compiler
- d. Firewall

MCQ 5. Prototype before code when:

- a. The question is "do people understand the idea?"
- b. You already have 1 million users
- c. The law froze the spec like waterfall NASA
- d. You are only drawing NAND

MCQ 6. Throwing a prototype away can be:

- a. Success (you learned cheaply)
- b. Shame
- c. A SQL error
- d. A median

Extra short questions

1. Define prototype from the book.
2. Three purposes.
3. What it is not.
4. A non-app prototype.

Extra long question

1. Design a paper prototype of a "lost-and-found" college app: 3 screens, 3 things you would watch in a test. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Early model of a product/service/app that shows how it may look or work. Not the final thing.
2. Communicate idea, test with users, find problems early.
3. Not the finished product; not necessarily coded; not an MVP (next topics).
4. A cardboard token for a queue system.

Long

1. Home, report item, search. Watch: do they find report in 30s, do they understand photos, do they trust privacy. Don't code yet.

Next lecture: 6.5 Types of Prototypes (Low, Mid, High-Fidelity)